UNIFIED MECHANICS

Current Scientific Position

Authority, closure, reproducibility, and the active frontier

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GOVERNING READER'S GUIDE TO THE UNIFIED MECHANICS CORPUS

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Abstract

Unified Mechanics is a theory of everything in the precise sense asserted by its corpus: it proposes one generative structure beneath general relativity, quantum field theory, and quantum mechanics, and it recovers their successful domains through a common carrier, action, matter, and readout architecture. This document states the current scientific position after reconciliation of the five-volume series, the companion derivation dossier, the Formal Register, the later matter and action results, the reproducibility layer, and the Nine-Step Closure. It distinguishes the status of the theory from the status of individual statements inside it. Exact derivations, conditional physical realizations, numerical certificates, and prospective measurements remain separately graded. The active frontier is finite and localized: a stationary mass selector, a microscopic matter-to-cell identity, a short-range response kernel, threshold and normalization calculations, precision likelihood work, and prospective physical measurements.

Reading contract

This guide applies four rules.

1. *The theory is not graded by averaging its sentences.* A proved identity, a conditional realization, and a future measurement may coexist in one chapter without acquiring one vague collective status.
2. *Authority follows explicit supersession.* Earlier statements remain provenance; later statements govern when they supply the named theorem, construction, computation, or certificate.
3. *Interpretation is not contradiction.* A non-unique realization remains under determination unless an incompatibility or empirical exclusion is established under declared premises.
4. *No measurement is written into existence.* A prospective map may be derived and frozen; apparatus data remain a physical act.

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CHAPTER 1

The current claim

1.1 What Unified Mechanics is

Unified Mechanics begins from recoverable distinction and the scale-stable two-channel relation

$$\varphi^2 = \varphi + 1, \qquad r = \frac{1}{2\varphi}, \qquad u = 1 - r. \tag{1.1}$$

The corpus develops these relations into a typed record calculus, finite carrier geometry, admissible reduction, local action, chiral matter, cosmological evolution, observer-relative readout, and an empirical protocol. The claim is not that established physics is discarded. The claim is that the successful domains of general relativity, quantum field theory, and quantum mechanics are effective expressions of this common lower structural layer.

> STATUS OF THE THEORY
>
> *Unified Mechanics is treated throughout the current edition as a theory of everything. Remaining numerical solutions, constitutive maps, matching conditions, and prospective measurements define its frontier. They do not convert the object into a merely aspirational programme.*

1.2 What scientific closure means here

Closure does not require every low-energy number to be a one-line algebraic constant. It requires a connected construction, explicit assumptions, common limits, defined observables, and a finite record of inputs that have not been derived. Let $\mathscr{G}$ be the corpus dependency graph. For every empirical sector p, the closure requirement is

$$\forall p \in \mathscr{P}_{\mathrm{emp}}, \qquad \exists \gamma : \mathscr{F}_{\mathrm{root}} \rightsquigarrow p, \qquad A(\gamma)\ \text{explicit}. \tag{1.2}$$

Reachability and compatibility establish the architecture. Uniqueness of a microscopic realization or numerical selector is a stronger local condition and is recorded at that edge.

CHAPTER 2

Chronology and authority

2.1 The dependency chronology

The corpus is read in the order in which physical commitment increases:

$$\begin{aligned}&\text{distinction and retained identity} \longrightarrow \text{ordered record}\\ &\longrightarrow \text{carrier and admissible reduction} \longrightarrow \text{action and matter}\\ &\longrightarrow \text{gravitation and cosmology} \longrightarrow \text{observation and registered evidence.}\end{aligned} \tag{2.1}$$

The five volumes follow this dependency order. The Derivation Programme traces the same structure across the conventional unification map. The Nine-Step Closure then reconciles the later theorems, computations, and certificates against the earlier editorial work queue.

2.2 Governing order

Where two statements differ because later work supplies an earlier missing construction, the current grade follows this order:

1. the current Core Freeze Manifest and Formal Register;
2. the core books;
3. issued technical papers in issue order;
4. reproducibility certificates;
5. later addenda and Helical Relativity records;
6. the Nine-Step companion register NS1–NS9;
7. historical editorial statements retained for provenance.

Chronology rule

If a later statement explicitly supplies the premise, construction, or certificate named as absent by an earlier statement, the earlier statement remains a correct record of its date while the later statement determines the current scientific grade. No mathematical conclusion is changed by this rule; it prevents a superseded absence from being reported as a present one.

CHAPTER 3

The continuous construction

3.1 Foundation, carrier, and action

The distinction-record language has a typed form and a vocabulary-invariance theorem. The rank-eight carrier, finite reduction, survivor algebra, matter module, family structure, and chirality bridge have declared hypotheses and finite certificates. The weighted square yields the registered Einstein–Cartan equations; the cluster pairing yields the gauge and chiral equations; minimal coupling yields the helical scalar source projection.

3.2 Matter and running

The chiral 16, Standard-Model charges, anomaly sums, and one-loop beta coefficients are fixed. The mass-squared operator lies in the boundary 15. The leading family-degenerate sector is therefore not an unstructured absence: the observed splitting is localized to a twelve-component stationary selector with a declared operator and boundary.

3.3 Cosmology and transfer

Renewal yields vacuum stress

$$T_{\Lambda}^{\mu\nu} = -\rho_{\Lambda} g^{\mu\nu}, \tag{3.1}$$

while conserved neutral retention yields dust at the effective level. Their scalar perturbations enter the standard transfer hierarchy with Unified Mechanics initial data. CAMB 2.0.1 replaces the earlier Saha-only estimate with non-equilibrium recombination and complete linear Einstein–Boltzmann evolution. In the frozen boundary realization the recorded outputs are

$$z_{*} = 1089.746, \quad r_{*} = 143.971\,\mathrm{Mpc}, \quad H_0 = 67.753\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{3.2}$$

with temperature peaks at $\ell = 220, 536, 813, 1126$.

3.4 Reproducibility and observables

The isolated rebuild regenerates all twenty-two corpus certificates. Closure count, Hubble basins, vacuum response, and short-range scale have explicit observable maps; the response kernel remains the named prerequisite for the short-range realization. Computational transfer has been executed. Physical protocols remain timestamped and outcome-neutral until prospective data exist.

CHAPTER 4

The active frontier

4.1 Primary obligations

Interface	Selecting work
Mass spectrum	Solve the twelve-component stationary boundary selector and connect its eigenvalues to the absolute scale.
Matter microphysics	Complete the discrete matter-to-cell constitutive identity beneath the effective covariant source.
Short-range response	Derive the granularity kernel that converts the fixed scale into a force, response, or null prediction.
Gauge matching	Compute the carrier threshold spectrum and overall normalization before precision endpoint comparison.
Cosmological inference	Evaluate the public likelihood and perform an independent HyRec/CosmoRec cross-run.
Physical evidence	Execute the preregistered apparatus and retention measurements without post-unblinding convention changes.

THE NARROW FRONTIER

These obligations are localized calculations and measurements at named interfaces. None licenses the algebraic, geometric, action, matter, cosmological, or reproducibility architecture to be described as absent.

CHAPTER 5

Canonical reading order

1. *Current Scientific Position* – the governing status and reader's contract;
2. *Complete Series* or Volumes I–V – the chronological theory;
3. *Derivation Programme* – the conventional unification map traced from inside the theory;
4. *The Nine-Step Closure* – later completion, computation, companion register, and prospective protocols;
5. the audit and source package – machine-readable provenance, certificates, scripts, and build sources.

The omnibus is an integrated edition, not a page-for-page concatenation of the standalone books. The complete source-disposition crosswalk is a supplementary audit table rather than part of the printed Formal Register. Earlier layouts remain available in the archive and do not compete with the canonical editions.

CURRENT CORPUS POSITION

Unified Mechanics is a functioning unified physical theory with a closed central architecture, a separately graded evidential record, reproduced computational results, and a finite active frontier. Future register entries may strengthen, narrow, or exclude particular realizations under stated premises; they do not erase the derivational chain by which the theory reaches its present form.